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NOTES AND CORRESPONDENCE

A Box-Lagrangian Rain-Drop Scheme

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Abstract

A new rain-drop scheme named box-Lagrangian rain-drop scheme is developed. In the Eulerian rain-drop scheme, since the change of the mixing ratio of rainwater q during unit time is calculated from $\partial q/\partial t = V\partial q/\partial z$, the time step interval Δt is restricted by the Courant-Friedrichs-Lewy (CFL) condition for rain falling: $V\Delta t/\Delta z < 1$, where V is the terminal velocity of rain falling, Δz the vertical grid spacing. In the box-Lagrangian rain-drop scheme, the above numerical constraint can be relaxed by the following method. The bulk of rainwater in a vertical grid box is dropped while keeping V constant during a time step interval, and it is partitioned into grid boxes existing in the space where it is dropped.

When Δt is less than the critical value, Δt_c, which is calculated from the CFL condition (*i.e.*, $\Delta t_c = \Delta z/V$), the box-Lagrangian scheme coincides with the Eulerian scheme. Furthermore, even when Δt is several times larger than Δt_c, the box-Lagrangian scheme drops rainwater stably with sufficient accuracy. The box-Lagrangian scheme is effective especially when many vertical layers are employed in the lower part of the model domain to express the atmospheric boundary layer in detail.

1. Introduction

The development of numerical weather prediction (NWP) models with high-resolution has become increasingly feasible so that fine detailed prediction for precipitation and strong wind can be operationally carried out. Prior to its development, the prime precipitating scheme suitable for NWP models must be determined in addition to the dynamical framework. In 10~20 km grid-resolution models, Kato and Saito (1995) concluded that the drag effect of hydrostatic water loading is more important in the simulation of moist convection than the non-hydrostatic effect. This conclusion suggests that a prognostic explicit scheme of cloud water and rainwater should be adopted to evaluate the effect of hydrostatic water loading in preference to introducing non-hydrostatic models.

An explicit rainwater scheme was applied to a numerical model for the first time by Das (1964). Kessler (1969) collated the early treatments of rainwater generation processes, which included autoconversion, accretion and evaporation. In application to mesoscale models, a prognostic explicit scheme of cloud water and rainwater was used successfully by Yamasaki (1977), and Yamasaki (1986) and Zhang *et al.* (1988) coupled it with a convective parameterization. When a prognostic explicit scheme is employed in numerical models with a grid size larger than 10 km, the rain terminal velocity V other than the air horizontal velocity restricts the time step interval Δt. In those models, Δt must be chosen independently of horizontal grid spacing Δx to satisfy the Courant-Friedrichs-Lewy (CFL) condition for rain falling ($V\Delta t/\Delta z < 1$, where Δz is the vertical grid interval). Especially when many vertical layers are employed in the lower part of the domain to express the atmospheric boundary layer in detail, Δt has to be so reduced as to calculate rain falling stably. Actually, for 10~20 km grid-resolution models in which a prognostic explicit scheme is employed, Δt = 15~30 sec has been used (*e.g.*, Yamasaki, 1977, Kato and Saito, 1995).

In numerical models with a grid size larger than 10 km, when we use the time step interval calculated from the CFL condition for air advection (*i.e.*, $V_a\Delta t/\Delta x < 1$, where V_a is the absolute value of air horizontal velocity), we must modify a prognostic explicit scheme as follows. Only for dropping rainwater Δt is reduced to satisfy the CFL condition for rain falling, and/or rainwater in the atmospheric boundary layer is not predicted. The above modification needs more calculation time, and/or can not

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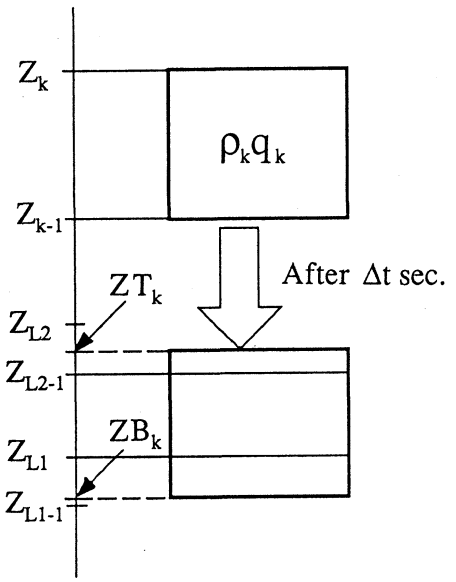

Fig. 1. The outline of the box-Lagrangian rain-drop scheme. ρ is the air density, q the mixing ratio of rainwater, Δt the time step interval, Z_k the top height of the k-th vertical grid box. ZT_k and ZB_k are respectively the heights of the top and the bottom of the k-th grid box after it is displaced downward during a time step interval.

simulate the atmospheric boundary layer successfully. In this paper, we introduce a new rain-drop scheme in which Δt is free from the CFL condition for rain falling.

2. Box-Lagrangian rain-drop scheme

A new rain-drop scheme named box-Lagrangian rain-drop scheme is described in this section. First, a rain-drop scheme is designed from the following equation when it is assumed anelastic.

$$\rho\frac{dq}{dt} = \rho\frac{\partial q}{\partial t} - \frac{\partial(\rho q V)}{\partial z} = 0. \tag{1}$$

Here, ρ is the air density, q the mixing ratio of rainwater and V the rain terminal velocity defined as $V = Aq^n$, where A and n are positive constants. In numerical models, the amount at the next time step is generally calculated from $\partial q/\partial t = \partial(rqV)/\rho\partial z$ (the Eulerian method), while we can consider a method which searches dropping points after a time step interval by using the left hand side of Eq. (1) (the Lagrangian method). The new rain-drop scheme is based on the Lagrangian method, but it drops not the rainwater at a vertical grid point but the bulk of rainwater in a vertical grid box.[1] Figure 1 shows the outline of the new scheme. It is designed so that the bulk of rainwater $\rho q\Delta z$ in a vertical grid box, which is defined by two vertical layers, is preserved while keeping V constant during a time step interval Δt. Here Δz is the depth of the layer. The k-th vertical grid box is displaced downward during a time step interval, and the heights of the top ZT_k, and the bottom ZB_k of the k-th grid box after displacement are determined first. Then the vertical layer where ZT_k, or ZB_k exists is searched;

$$ZT_k{}^m = Z_k - V_k{}^m\Delta t \subset [Z_{L2-1}, Z_{L2}]$$

and

$$ZB_k{}^m = Z_{k-1} - V_k{}^m\Delta t \subset [Z_{L1-1}, Z_{L1}], \tag{2}$$

where Z_k is the top height of the k-th vertical grid box, the superscript 'm' denotes the value at the time step 'm' and the brackets $[T, B]$ represent the interval in space between T and B.

Next, the dropped bulk of rainwater is partitioned into vertical grid boxes between ZT_k and ZB_k. When the displaced box interacts with the ground surface, the rainwater below it is removed from the model domain as precipitation. In the case of $L1 \neq L_2$:

$$\begin{aligned} P_{L1,k}{}^{m+1} &= \rho_k q_k{}^m(Z_{L1} - ZB_k), \\ P_{L,k}{}^{m+1} &= \rho_k q_k{}^m(Z_L - Z_{L-1}) \\ &\qquad \text{for } L1 < L < L2, \end{aligned}$$

and

$$P_{L2,k}{}^{m+1} = \rho_k q_k{}^m(ZT_k - Z_{L2-1}). \tag{3}$$

In the case of $L1 = L2$:

$$P_{L1,k}{}^{m+1} = \rho_k q_k{}^m(ZT_k - ZB_{k-1}), \tag{4}$$

where $P_{L,k}{}^{m+1}$ is the dropped rainwater in the L-th grid box at the next time step from the k-th grid box, and the subscript 'k' denotes the value at the k-th vertical grid point. After summing up the dropped rainwater from all vertical grid boxes, the mixing ratio of rainwater at the next time step is calculated as

$$q_L{}^{m+1} = \sum^{k} P_{L,k}{}^{m+1}/\rho_L(Z_L - Z_{L-1}). \tag{5}$$

When the CFL condition ($\Delta z > V\Delta t$) is satisfied, $L1 = k-1$ and $L2 = k$ are obtained from Eq. (2). By substituting $L1 = k-1$ and $L2 = k$ into Eqs. (3) and (5), the change of q during unit time, $(\partial q/\partial t)_k{}^m$, is obtained as

$$\begin{aligned} \left(\frac{\partial q}{\partial t}\right)_k^m &= \left\{\frac{P_{k,k+1}{}^{m+1} + P_{k,k}{}^{m+1}}{\rho_k(Z_k - Z_{k-1})} - q_k{}^m\right\}/\Delta t \\ &= \frac{\rho_{k+1}q_{k+1}{}^m V_{k+1}{}^m - \rho_k q_k{}^m V_k{}^m}{\rho_k(Z_k - Z_{k-1})}. \end{aligned} \tag{6}$$

1 A similar method to the box-Lagragian rain-drop scheme has been applied to gas dynamics (Leveque, 1988).

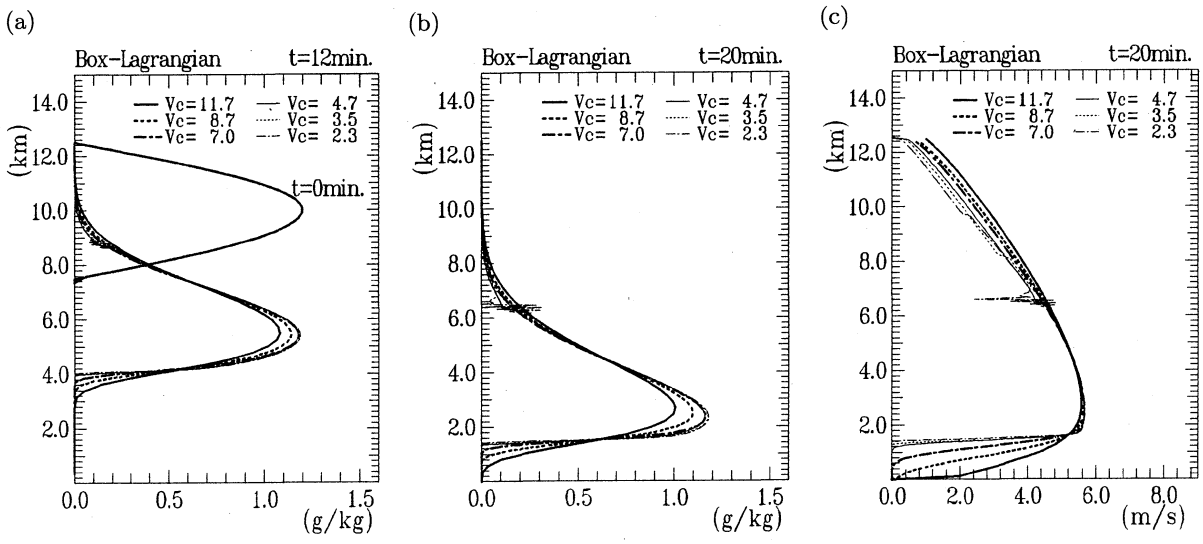

Fig. 2. The vertical profile of the mixing ratio of rainwater simulated by the box-Lagrangian rain-drop scheme at $t =$ (a) 12 and (b) 20 min. (c) That of the rain terminal velocity at $t = 20$ min.

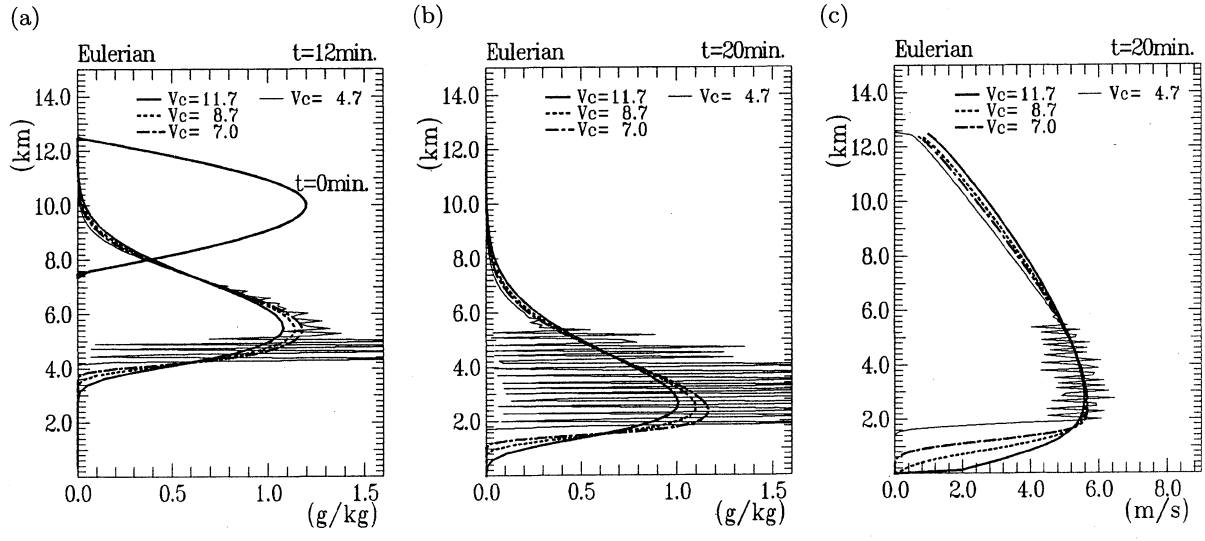

Fig. 3. As in Fig. 2, except for the Eulerian rain-drop scheme.

On the other hand, from Eq. (1) $(\partial q/\partial t)_k{}^m$ in the Eulerian rain-drop scheme is given as

$$\left(\frac{\partial q}{\partial t}\right)_k^m = \frac{\partial(\rho q V)_k{}^m}{\rho_k \partial z_k} = \frac{(\rho q V)_{k+1}{}^m - (\rho q V)_k{}^m}{\rho_k \Delta z_k}. \quad (7)$$

Thus, from Eqs. (6) and (7) both the box-Lagrangian and the Eulerian schemes coincide with each other when the CFL condition is satisfied.

3. One-dimensional validation test

To verify the numerical stability and accuracy of the box-Lagrangian rain-drop scheme, we examine its results against those simulated by the Eulerian rain-drop scheme. In our simulations, $V\Delta t/\Delta z$ is changed from $V\Delta t/\Delta z < 1$ (*i.e.*, the CFL condition for rain falling is satisfied) to $V\Delta t/\Delta z > 1$ (not satisfied).

3.1 Model and initial condition

In the one-dimensional simulation, in order to examine only the falling process of rain-drops, the air density is assumed constant and the water phase change is not considered. Here, V_c is defined as the ratio of Δz to Δt ($V_c = \Delta z/\Delta t$). Under the CFL condition for rain falling ($V_c > V$), V_c has to be chosen larger than the maximum of the rain terminal velocity $V_{\max}$. In the present study, $V_{\max}$ is considered to be 7.0 m/s and the numerical stability and accuracy of the box-Lagrangian rain-drop scheme is examined for $V_c = CV_{\max}$, where $C = 5/3$, 5/4, 1, 2/3, 1/2 and 1/3. Δt is fixed to be 10 sec, while

Δz and number of vertical layers are changed. For convenience, V is computed by the following simple formula which is proposed by Ogura and Takahashi (1971).

$$V = 3.12 \times 10^4 (\rho q)^{0.125}, \tag{8}$$

where $\rho = 1.0$ kg/m^3 is used.

The initial vertical profile of q is given as

$$\begin{aligned} q(z) &= 1.2 \times 10^{-3} \cos\left(\frac{\pi}{2} \times \frac{Zc - z}{Dz}\right) \\ &\qquad (Zc - Dz \le z \le Zc + Dz) \\ &= 0.0 \quad (z < Zc - Dz,\ Zc + Dz < z), \end{aligned} \tag{9}$$

where Zc and Dz are 1.0×10^4 m and 2.5×10^3 m, respectively.

3.2 Results

The vertical profiles of q and V simulated by the box-Lagrangian and the Eulerian rain-drop schemes are displayed in Figs. 2 and 3, respectively. Figure 4 shows the time series of precipitation intensity. From Figs. 2c and 3c, $V_{\max}$ is almost 6.0 m/s, so that for $V_c > 6.0$ m/s the CFL condition for rain falling is satisfied. The Eulerian scheme drops rainwater stably for $V_c > 6.0$ m/s, whereas the result for $V_c = 4.7$ m/s (< 6.0 m/s) suggests that the Eulerian scheme should not be used when the CFL condition is not satisfied (Figs. 3a and 3b). For $V_c > 6.0$ m/s, the results simulated by the box-Lagrangian and Eulerian schemes coincide with each other, which indicates that the numerical stability and accuracy of the box-Lagrangian scheme is the same as those of the Eulerian scheme under the CFL condition. Both in Figs. 2 and 3, the maximum of q becomes slightly smaller as V_c increases. This is caused by numerical error.[2]

Figures 2 and 4a show that the box-Lagrangian scheme drops rainwater stably with sufficient accuracy even for $V_c < 6.0$ m/s, and that the results almost agree with those for $V_c = 7.0$ m/s. Figure 2 also shows that the upper part of the profile fluctuates a little. However, this fluctuation barely influences results[3] (see Fig. 4a). Thus, in the box-Lagrangian scheme Δt can be several times larger than the critical value Δt_c under the CFL condition (*i.e.*, $\Delta t_c = \Delta z / V_{\max}$).

2 To reduce this numerical error, positive definite advection schemes (*e.g.*, Smolarkiewicz, 1983, Hue and Arakawa, 1990) have been developed. However Δt in these schemes is also restricted by the CFL condition for rain falling.

3 In numerical models to simulate atmospheric phenomena, the region where Δt is to be several times as large as the critical value under the CFL condition for rain falling ($= \Delta z / V$) is the lower part of the model domain, so that the influence of this small wave-like fluctuation on results is more insignificant.

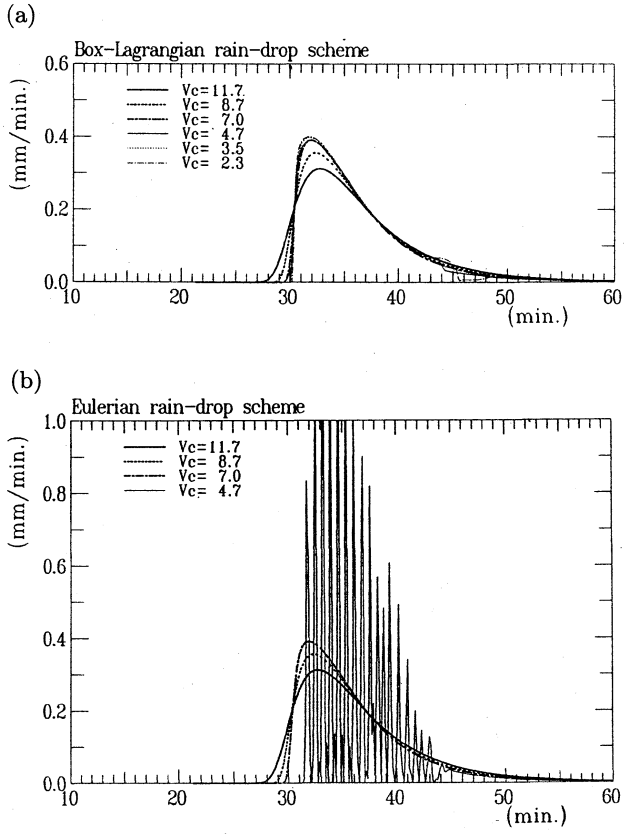

Fig. 4. Time series of precipitation intensity simulated by (a) the box-Lagrangian and (b) the Eulerian rain-drop scheme. $V_c = \Delta z / \Delta t$, where Δz is the vertical grid spacing and Δt the time step interval. The plots are truncated at the top bowndary of the graph box.

4. Summary and conclusions

In this paper a new rain-drop scheme named box-Lagrangian rain-drop scheme has been described. Under the CFL condition for rain falling ($\Delta t < \Delta z / V$), the new scheme coincides with the Eulerian rain-drop scheme. However, for $\Delta t > \Delta z / V$, the box-Lagrangian scheme can drop rainwater stably with sufficient accuracy while the Eulerian scheme becomes unavailable.

In 10~20 km grid-resolution models, Kato and Saito (1995) showed that the drag effect of hydrostatic water loading is important. To include this effect, we need to predict the amount of raindrops. In these models, the time step interval calculated from the CFL condition for air advection (*i.e.*, $\Delta t < \Delta x / V_a$, where V_a is the absolute value of air horizontal velocity) is about 30~60 sec. For this time step interval, if Δt is restricted by the CFL condition for rain falling, the vertical grid interval needs to be larger than 180~360 m when the maximum of the rain terminal velocity $V_{\max}$ is

6.0 m/s. However, the smallest vertical grid interval in NWP models is shorter than 100 m because many layers are employed in the lower part of the model domain to model the atmospheric boundary layer in detail. The box-Lagrangian scheme enables $\Delta t = 30\sim60$ sec for 10~20 km grid-resolution models, because Δt can be several times greater than the critical value Δt_c under the CFL condition for rain falling (*i.e.*, $\Delta t_c = \Delta z/V_{\max}$). The box-Lagrangian scheme being employed, we can introduce the drag effect into mesoscale models without reducing Δt and thereby increasing calculation time.

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ボックスラグランジアン雨滴落下スキーム

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新たな雨滴落下スキーム (box-Lagrangian スキーム) を開発した。従来の Euler 型のスキームでは、雨滴の混合比 qの時間変化を$\partial q/\partial t = V\partial q/\partial z$から計算しているので、タイムステップ$\Delta t$ を雨滴の落下に対する CFL 条件 ($V\Delta t/\Delta z < 1$、ここで Vは雨滴の落下速度、Δzは鉛直の格子間隔) をも満足するように決定しなければならない。そこで、新しいスキームでは 1 鉛直格子箱にある雨滴の総量 (可降水量) が完全に保存するように雨滴の落下を考え、そうすることにより雨滴の落下に対するΔt の束縛条件を取り除くことができた。すなわち、可降水量を一定の Vで落下させ、Δt 後の落下位置にある格子に配分する方法である。

雨滴の落下に対する CFL 条件から考えられる最大のタイムステップ$\Delta t_c(= \Delta z/V)$ よりも小さいΔt に対しては box-Lagrangian スキームは Euler 型のスキームと一致する。さらに、Δt をΔt_cの数倍にした場合でも box-Lagrangian スキームは精度良く安定に雨滴の落下を計算した。数値モデルの下部の格子間隔を大気境界層を表現するために細かく取る場合、box-Lagrangian スキームは特に有効な手段となる。